

MASON MILLER

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EDUCATION

University of Michigan

Bachelor of Science in Computer Science

Ann Arbor, MI

August 2022 – May 2026

- **GPA:** 3.9/4.0 | **Awards:** James B. Angell Scholar, William J. Branstrom Prize, University Honors (all terms)

TECHNICAL SKILLS

Languages: C++, C, Go, Python, Bash, SQL

Systems: Linux, XNU, POSIX, pthreads, virtual memory, mmap, custom allocators (dldmalloc), io_uring, libaio, SPDK, NVMe

Distributed & Tools: Paxos, sharding, consistency models, gRPC, TCP/IP, RocksDB, SQL Server, GDB, Valgrind, ASan, QEMU, Docker, Kubernetes, GCP, AWS, Claude Code

EXPERIENCE

Citadel

Software Engineer Intern – Electronic Trading Engineering Core

New York, NY

June 2026 – August 2026

- **Internal Data Platform:** Built a distributed service exposing shared relational data to trading applications on managed container infrastructure; isolated tenants via namespaced metadata with per-dataset access control, kept storage append-only for full revision history, and gated edits behind multi-party approval
- Generated typed client bindings so consuming applications read data natively, with a streaming channel for incremental updates
- **Application State Persistence (prototype):** Designed a library restoring in-memory state from a persisted snapshot rather than replaying a week-long event log after a crash, serving allocations from a fixed pre-mapped arena behind a custom persistent allocator so pointers stay valid across restarts
- Snapshotted without stalling the hot path by capturing a consistent read-only view of memory and writing it asynchronously, validating restores against integrity and version metadata

Ordered Systems Lab

Research Assistant (Advisor: Prof. Ryan Huang)

University of Michigan

May 2024 – May 2026

- Joined as the lab's first sophomore member; curated benchmarks and built test harnesses for a coding agent triaging complex bugs in large open-source distributed systems (ZooKeeper, HDFS, Kafka), annotating real-world failures across three research efforts
- Built Docker-based testing infrastructure and context-injection pipelines for Cirdan, an LLM distributed-systems debugger; separately benchmarked async I/O libraries (libaio, liburing) against RocksDB across queue depths and workload patterns

Apple CoreOS

Software Engineer Intern – Storage Technologies

Cupertino, CA

May 2025 – August 2025

- Enabled user-space testing of kernel storage drivers without physical hardware probing by building a C++ tool that fakes the XNU IOKit framework, with a fake storage device, automatic driver matching, and error injection into mock block devices
- Raised driver code coverage by over 80% with the new harness and used it to expose a 15-year-old bug in the XNU kernel

Qumulo

Member of Technical Staff Intern – Cloud Read Cache

Seattle, WA

January 2025 – April 2025

- Cut minimum cluster cost by 33% by designing and deploying single-node clusters with disk-level fault tolerance, and built automated AWS/Azure performance testing to benchmark single- vs. multi-node configurations
- Led a distributed key/value read cache upgrade with safe key remapping, quorum barriers, and RPC-based shard coordination; instrumented cache hits, misses, and gap-fills to optimize the read path

EECS Staff, University of Michigan

Instructional Assistant – Distributed Systems (W26), Operating Systems (F25)

Ann Arbor, MI

August 2025 – May 2026

- Taught and held office hours on Paxos consensus, replicated state machines, primary-backup replication, consistency models, and sharding (EECS 491); threads, synchronization, scheduling, virtual memory, and file systems (EECS 482), providing 7-day forum coverage and 5–10 office hours weekly

PROJECTS

Sharded Key/Value Service with Paxos Groups (Go)

- Built a fault-tolerant sharded key/value store with shard migration across Paxos-replicated groups, a shard master for configuration and load balancing, and atomic two-phase handoff that guarantees linearizability across reconfiguration

Multi-threaded Network File Server (C/C++)

- Built a concurrent network file server over TCP serving many simultaneous users across nested directories, using fine-grained reader-writer locking with hand-over-hand traversal and ordered disk writes preserving metadata invariants

Virtual Memory Pager (C/C++)

- Designed a demand-paged virtual memory system for multiple processes with swap- and file-backed pages, clock replacement with eager swap reservation, and copy-on-write `fork()` with deferred fault handling to minimize disk I/O and page copies

Preemptive Thread Library (C/C++)

- Developed a preemptive kernel-level thread library for multiple CPUs with timer interrupts, `ucontext` context switching, mutexes and condition variables under correct interrupt management, and FIFO scheduling with CPU suspend/resume